



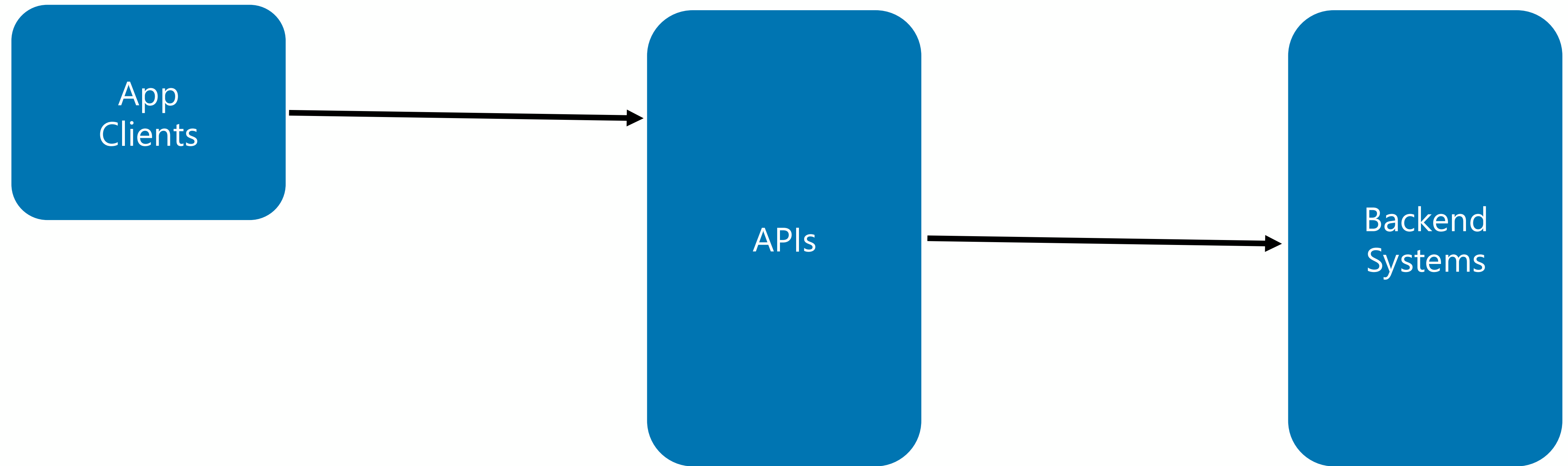
TRAILHEAD
TECHNOLOGY PARTNERS

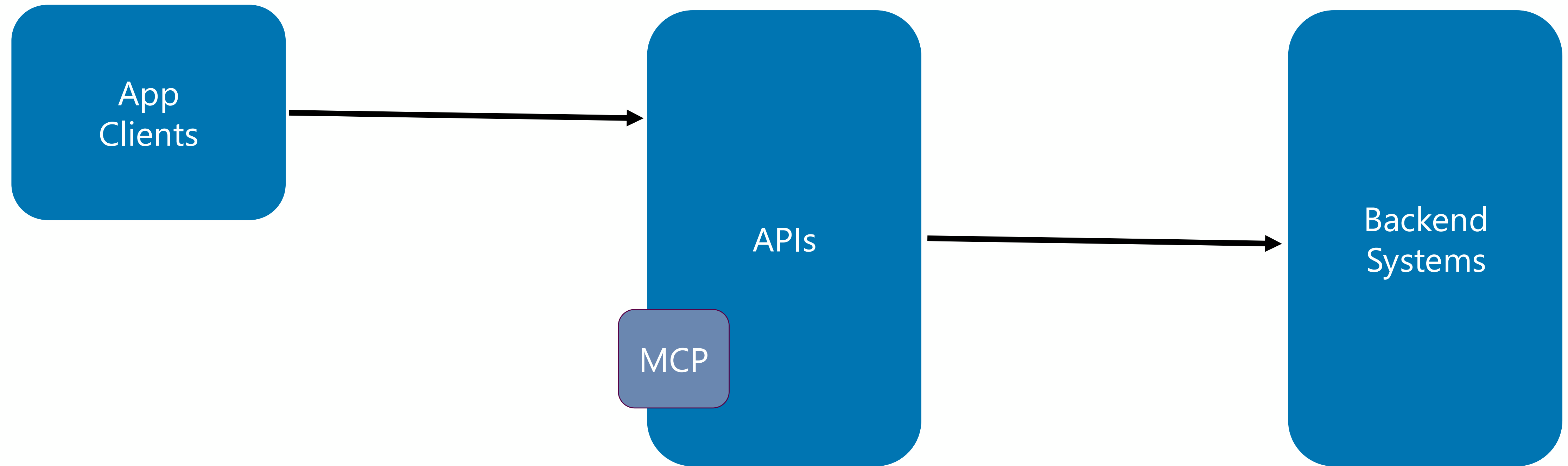
Old API, New Tricks

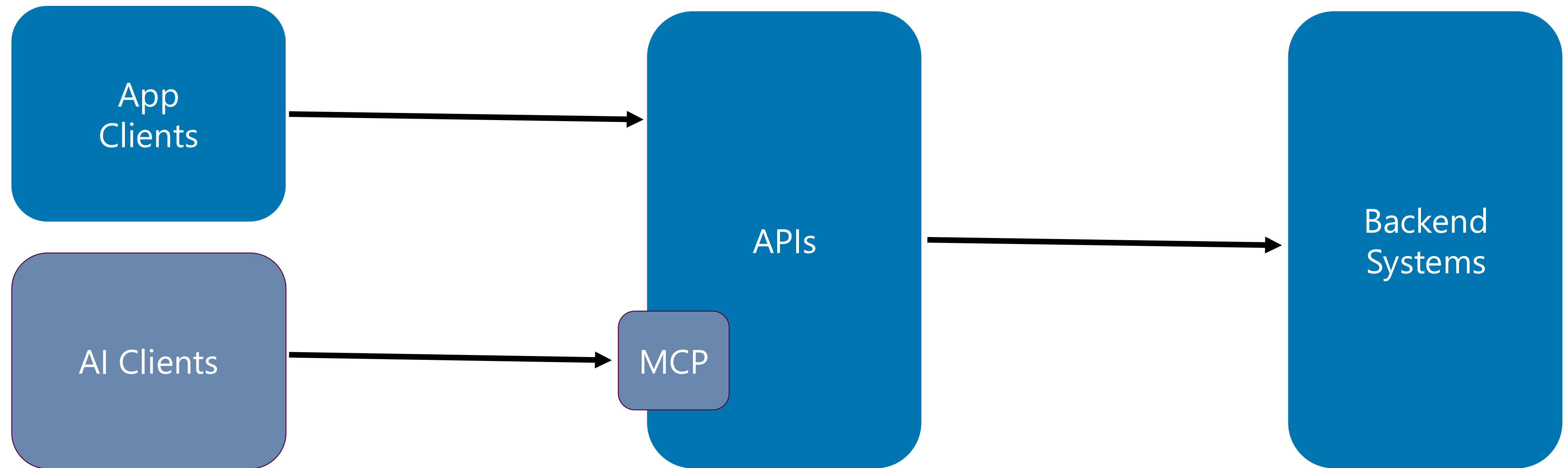
Add MCP to Existing .NET
REST Endpoints



Jonathan "J." Tower







MCP Usage Scenarios

Sales assistant updating CRM, scheduling follow-ups, crafting briefs

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Scheduling assistant, coordinating calendars, proposing times, sending invites

Preview

- 01** What is MCP?
- 02** MCP vs REST
- 03** MCP C# SDK
- 04** Demo + Tool Design
- 05** Debugging & Security

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Founding Partner

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- VP of .NET Foundation
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- trailheadtechnology.com/blog
- [jtowermi](#)
- Jonathan "J." Tower

LET'S TALK



tinyurl.com/th-offer

github.com/trailheadtechnology/dotnet-mcp-rest

Introduction to MCP

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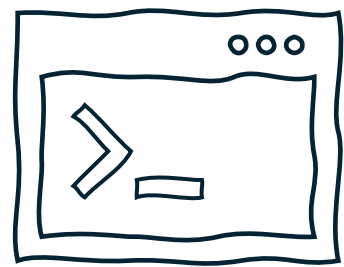
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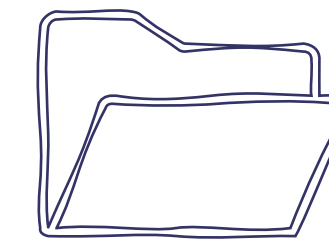
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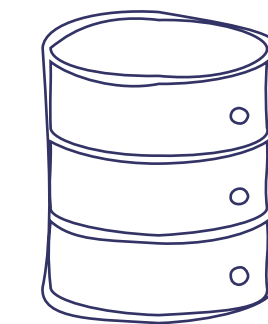
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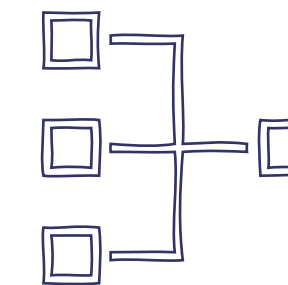
LLM



File System

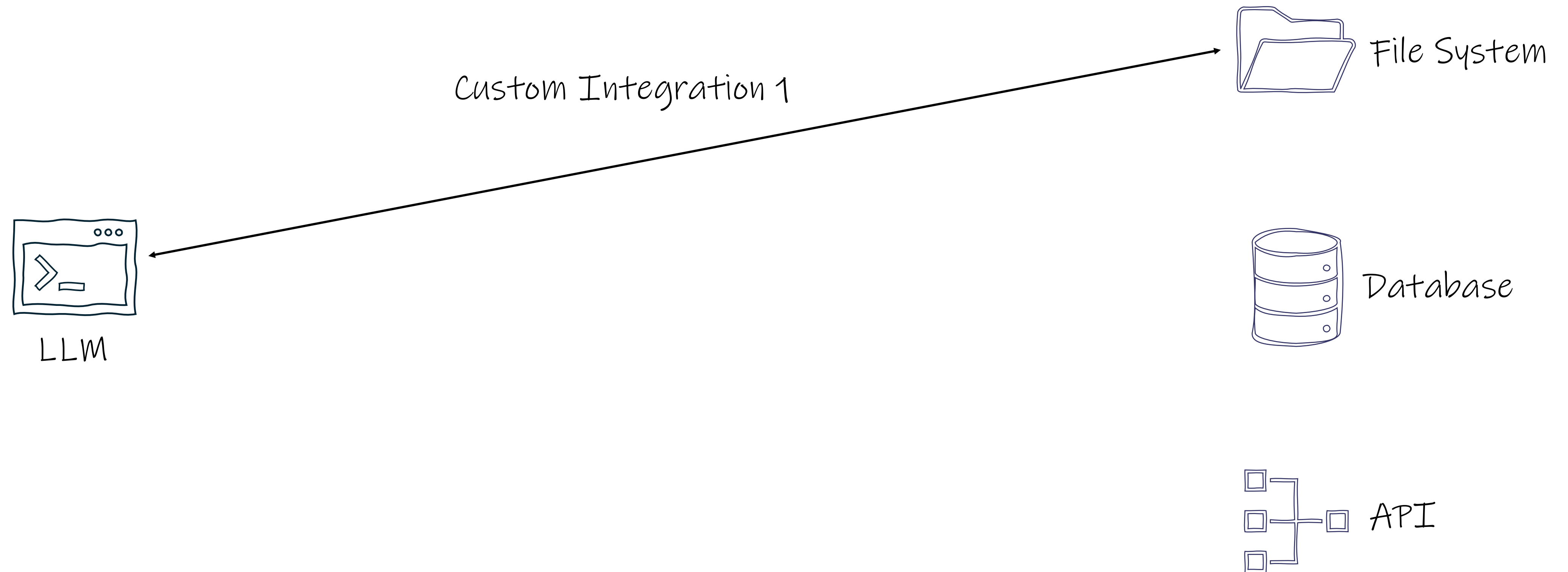


Database

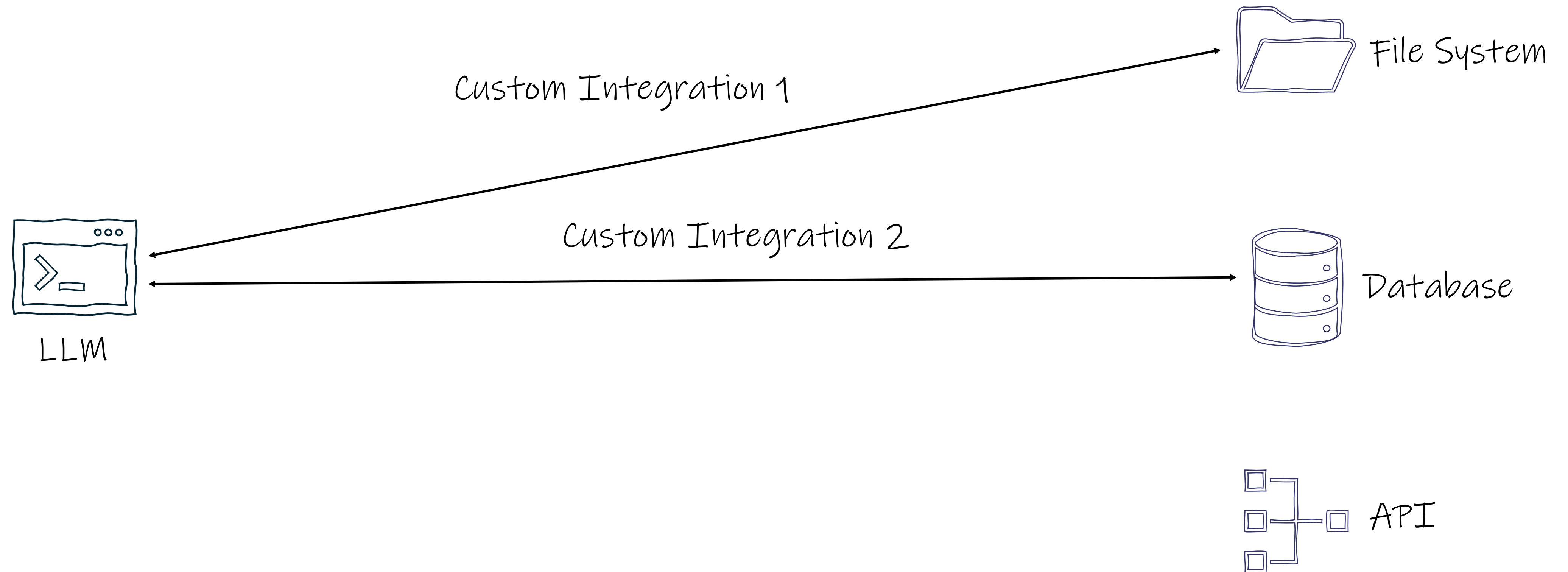


API

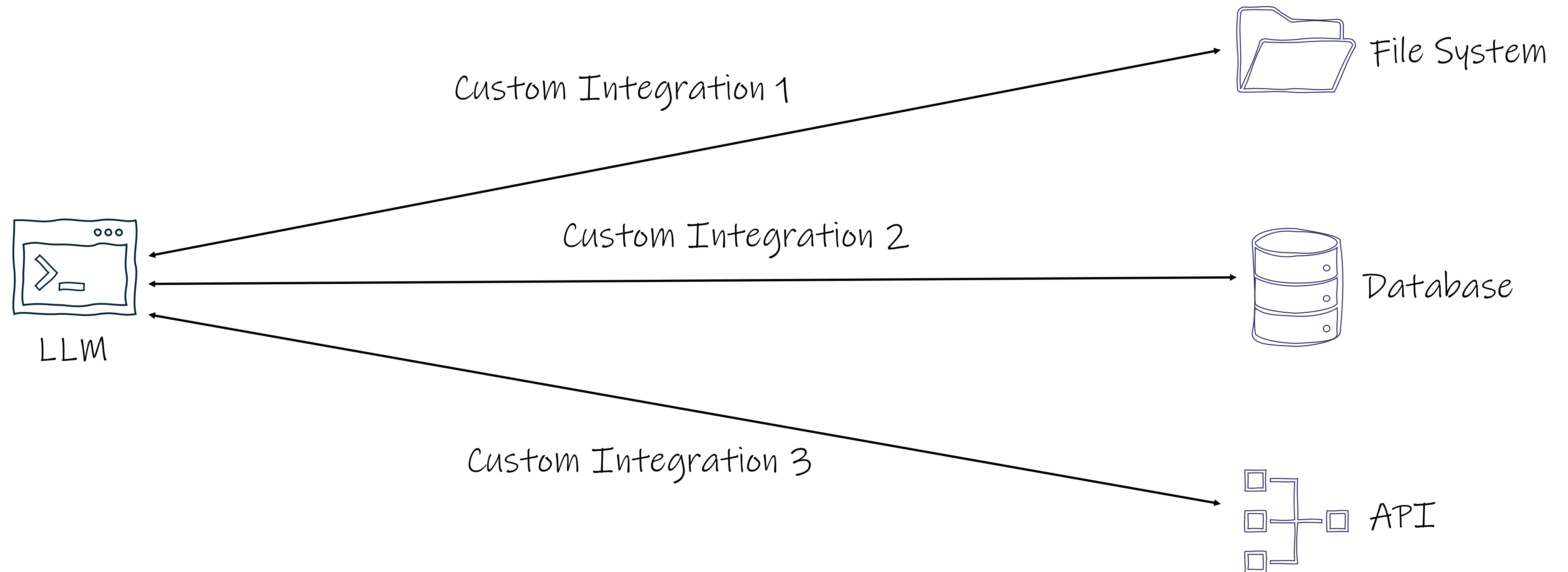
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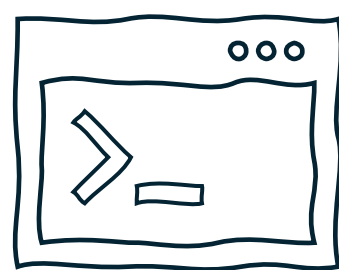
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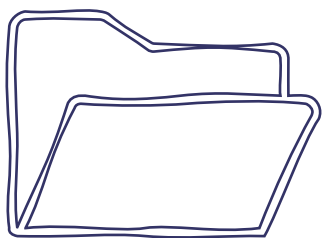


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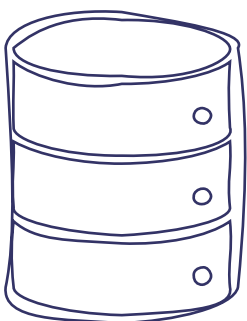


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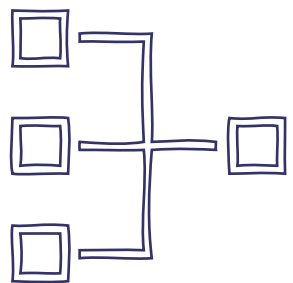
MCP Protocol



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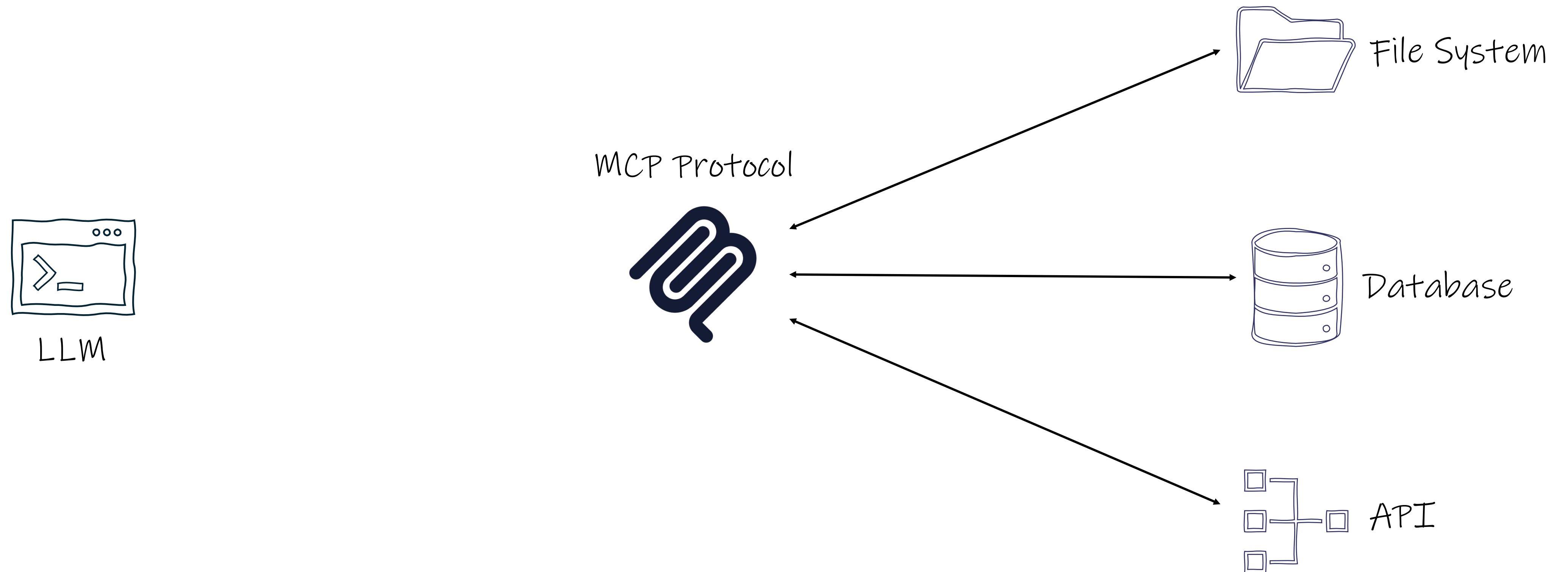


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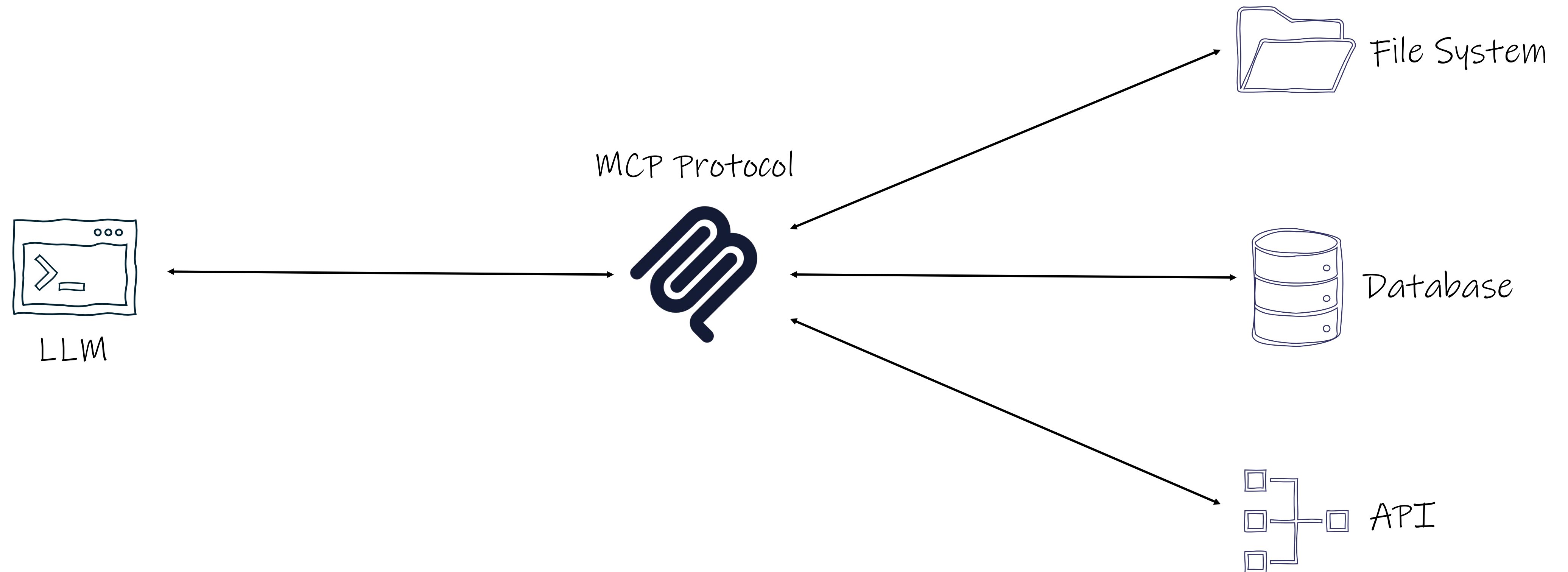


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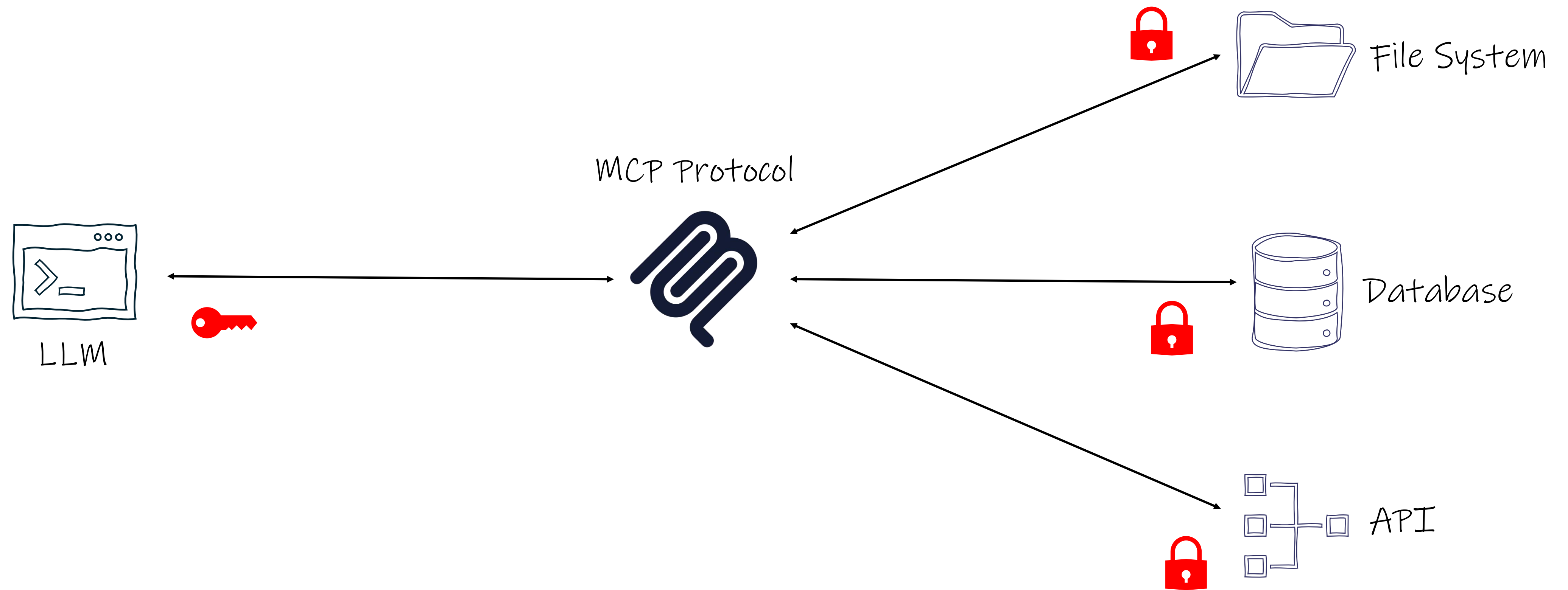
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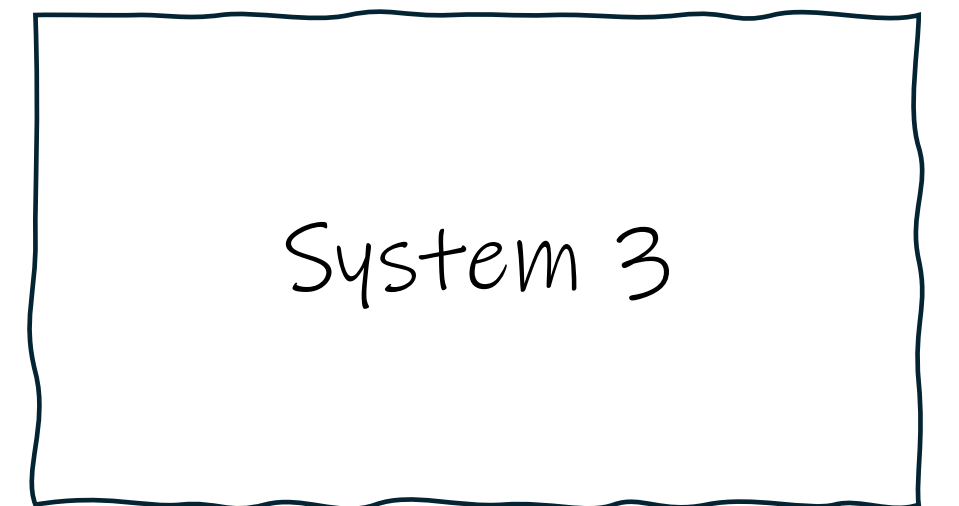
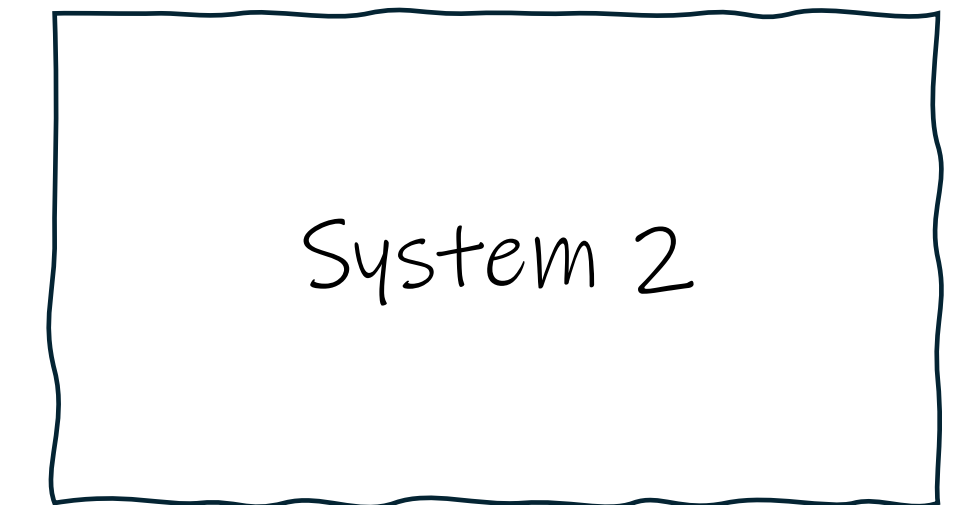
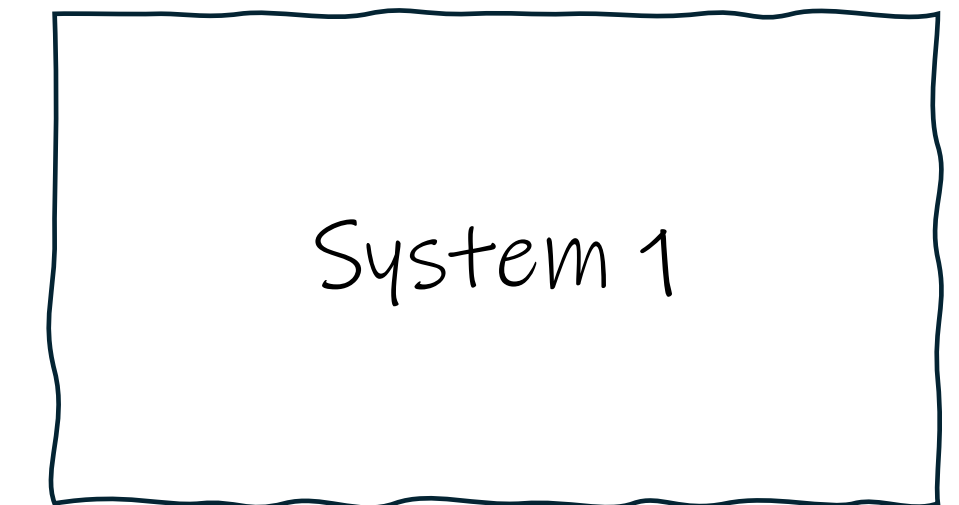
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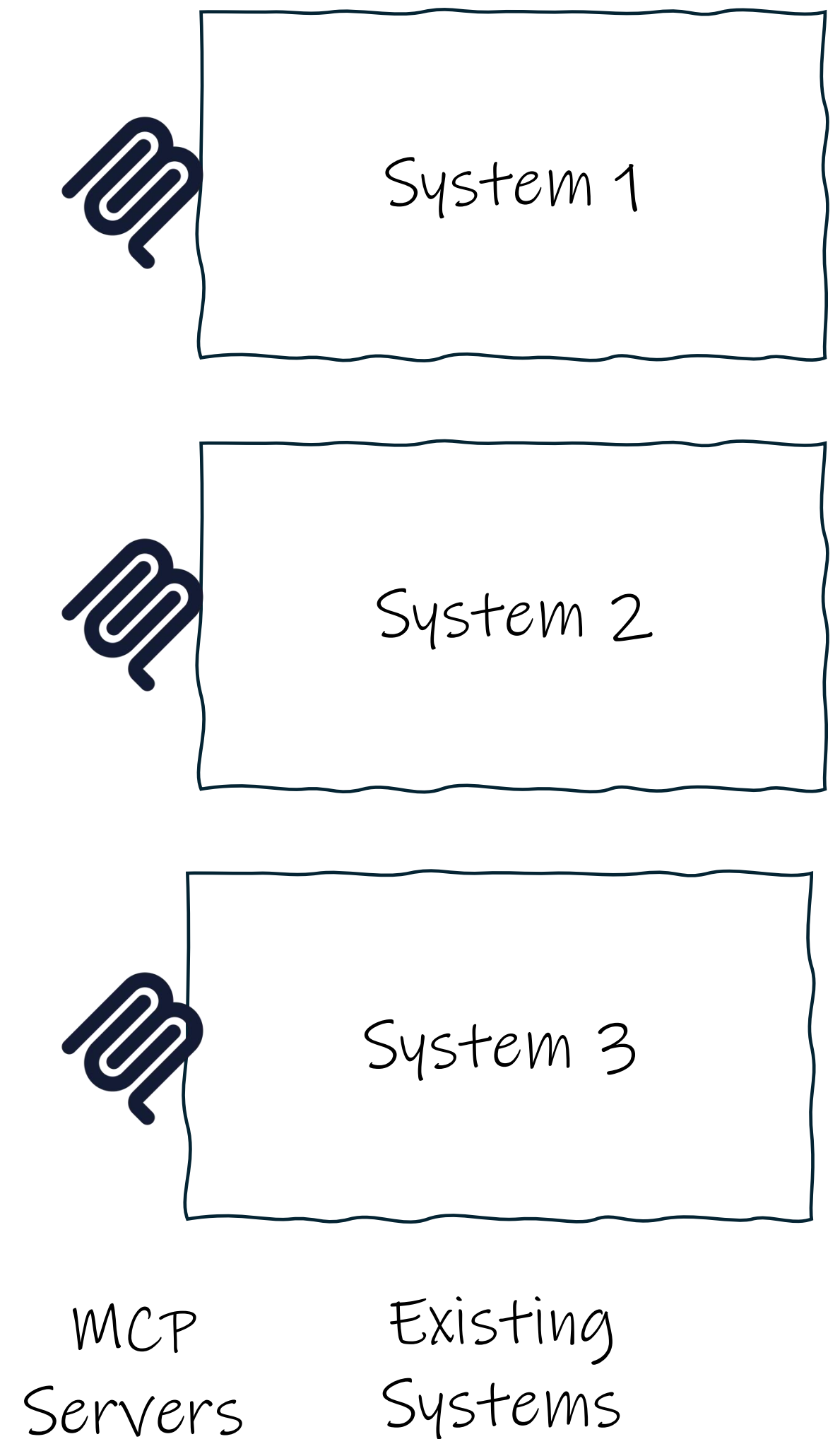


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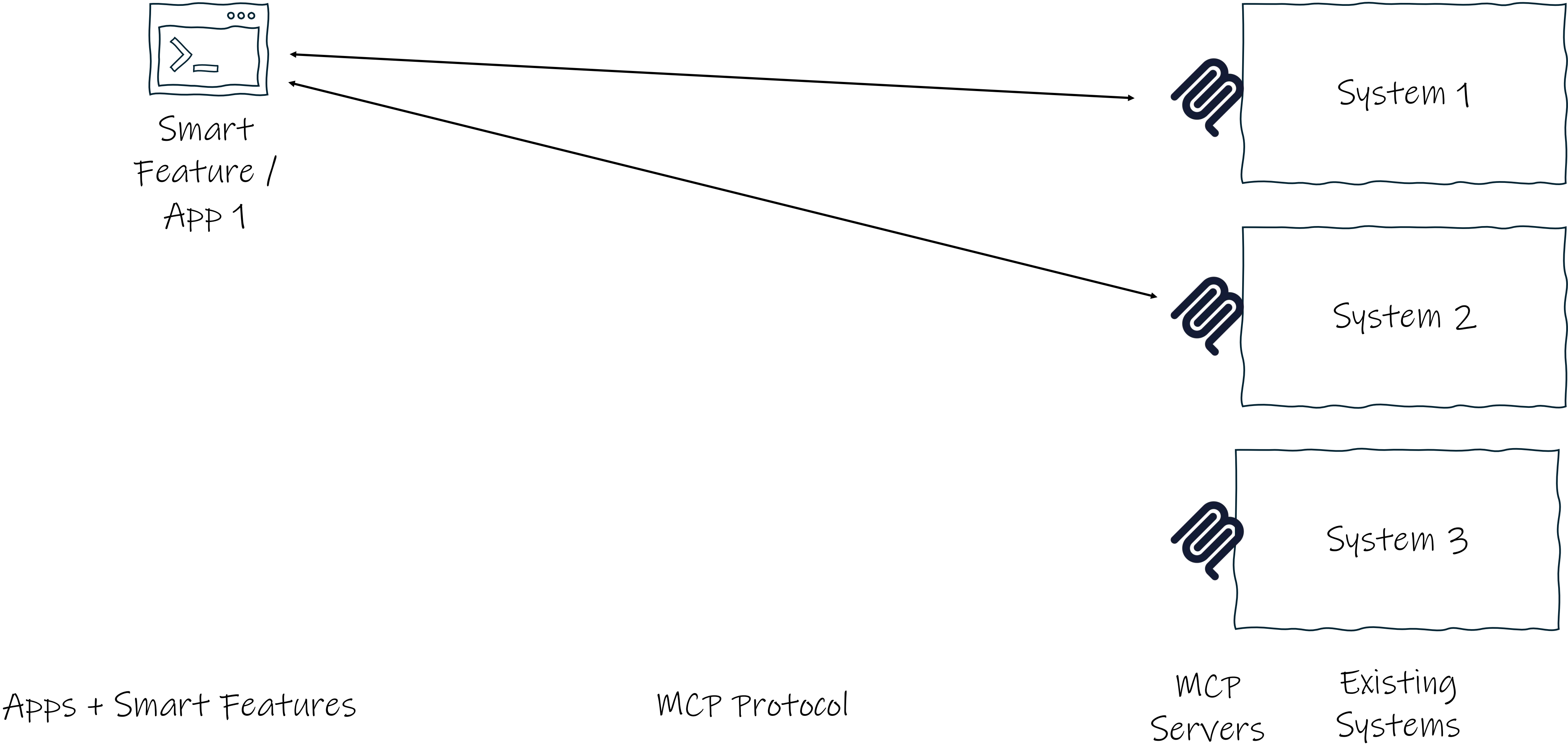


Existing
Systems

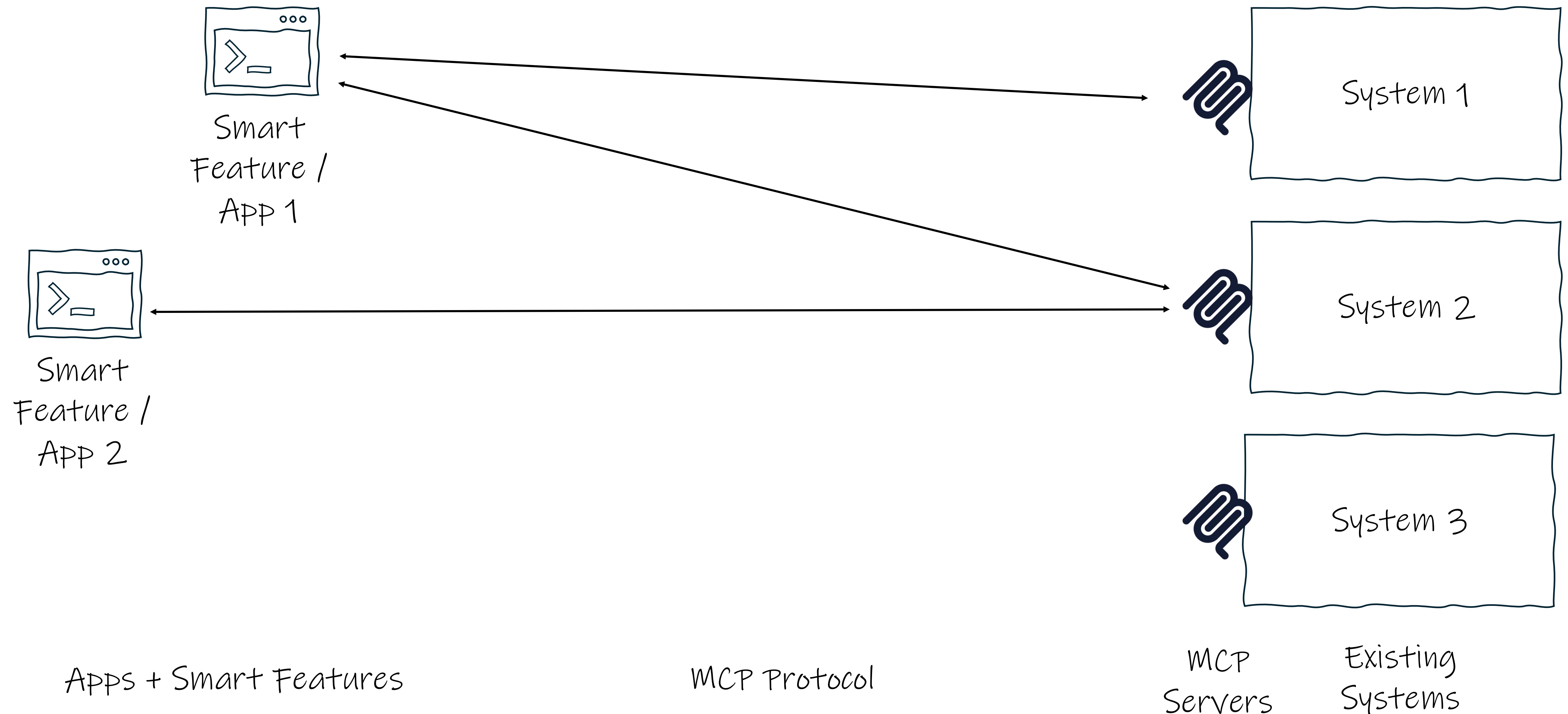
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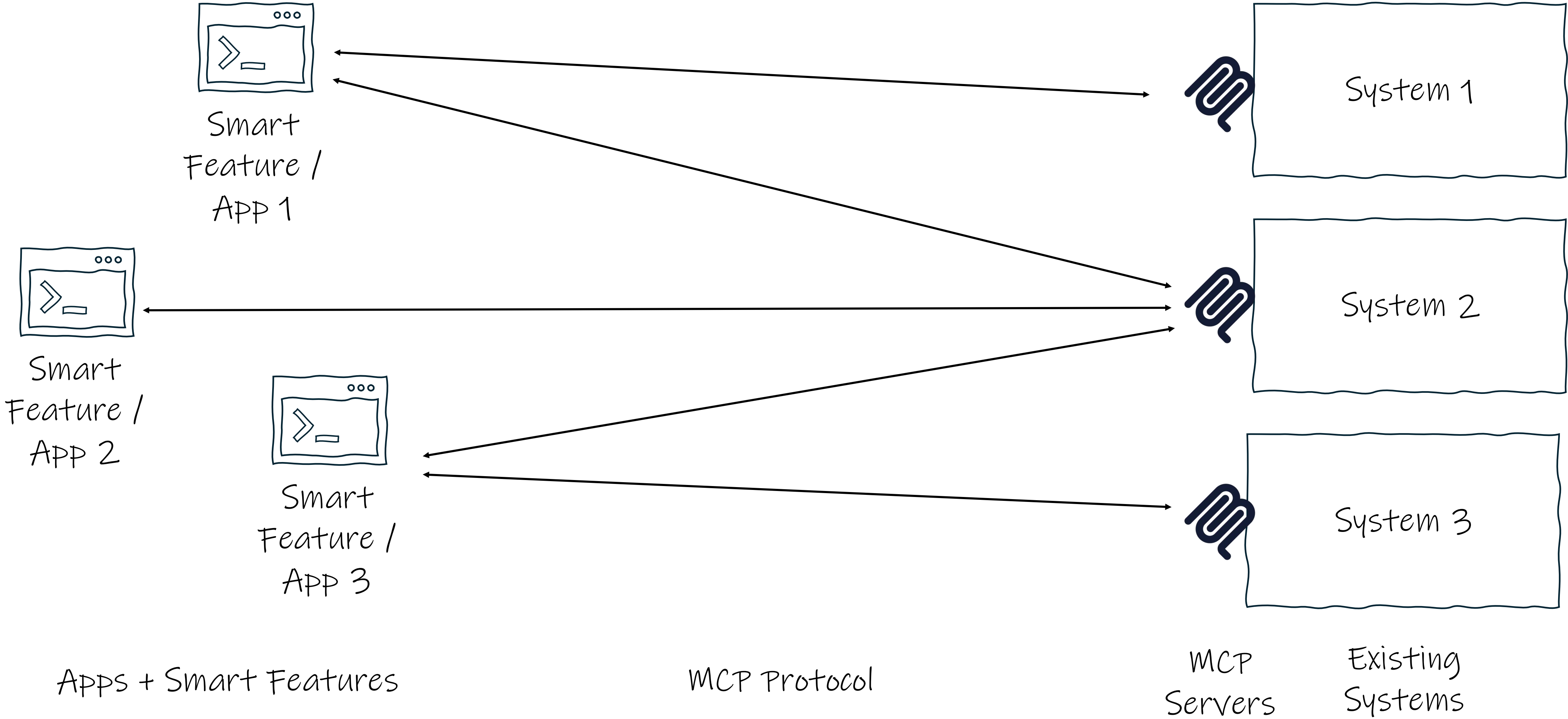
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MCP Transports

stdio

Local process over stdin/stout; no network, fast and simple.

Streamable HTTP

HTTP POST in, SSE stream out; remote-friendly, works with auth and proxies (formerly SSE)

Custom

Spec notes custom transports but not standardized

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REST vs MCP

REST vs MCP

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Primary audience	Other developers & services	LLMs/agents (plus humans)
Discovery	OpenAPI/docs (manual; static)	Runtime discovery (list_tools/resources/prompts)
Contract	URIs + HTTP verbs + status codes	Named tools + JSON Schemas + structured results
Invocation & transport	Request/response over HTTP	Per-call tool exec via stdio or streamable HTTP
Orchestration	Client stitches multiple calls	Agent plans multi-step tool chains
Streaming	Ad hoc/rare	First-class partial/progressive outputs
Auth model	HTTP auth per request	Transport-level; server injects downstream auth
Best use	BE integration APIs	Quick AI integration over existing endpoints

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MCP clients and servers

Build clients and servers in minutes using the SDK's ready-made client and server primitives.

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Attribute-based discovery

Attribute-based auto-discovery exposes your methods as MCP tools and prompts with a simple scan.

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MEAI integration

Microsoft.Extensions.AI native. Plug into DI, hosting, and logging you already use in .NET.

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Transports included

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Bridges to chat stacks

Chat/agent bridges. Surface tools as callable functions in chat client stacks.

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v2.0

Realigned to the latest spec: discovery-first negotiation with automatic fallback to the legacy initialize handshake.

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v2.1

A subscriptions/listen handler lets clients receive live updates when server resources change.

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HTTP serving is now stateless by default — simpler scale-out and load balancing for production hosting.

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Improved HTTP transport negotiation and fallback so clients connect reliably across protocol revisions.

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Hybrid Session Mode

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New `HttpServerSessionMode` lets one endpoint serve both older stateful clients and newer stateless clients during migration.

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Identity Assertion Auth Grant

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Support for the Identity Assertion Authorization Grant plus security hardening for enterprise auth flows.

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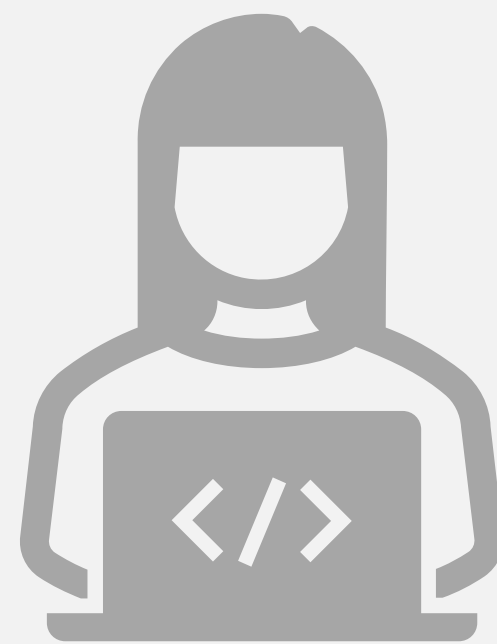
Tool Design Quality

Descriptions Are Your API Contract

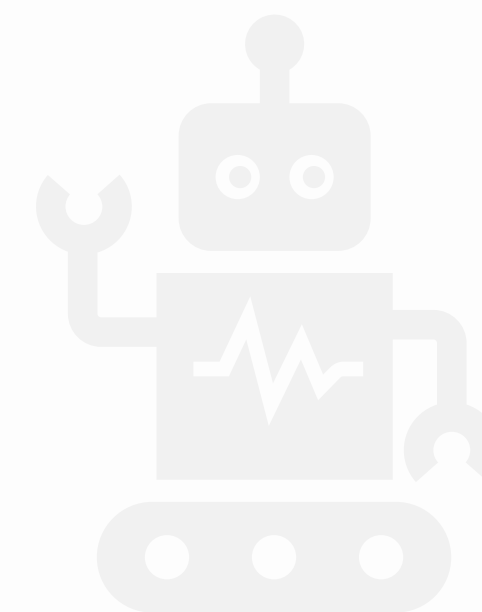
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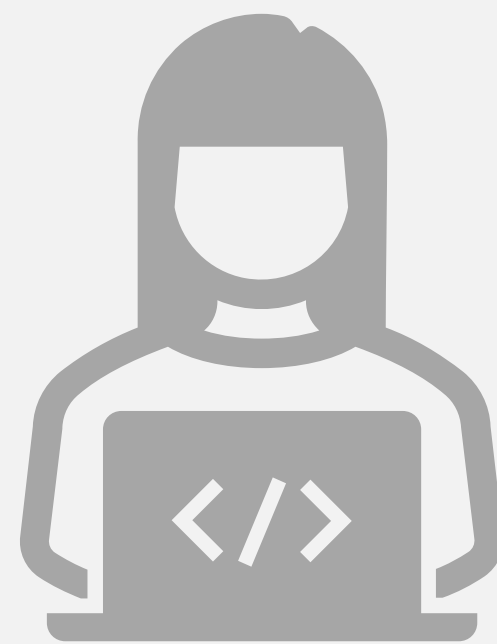


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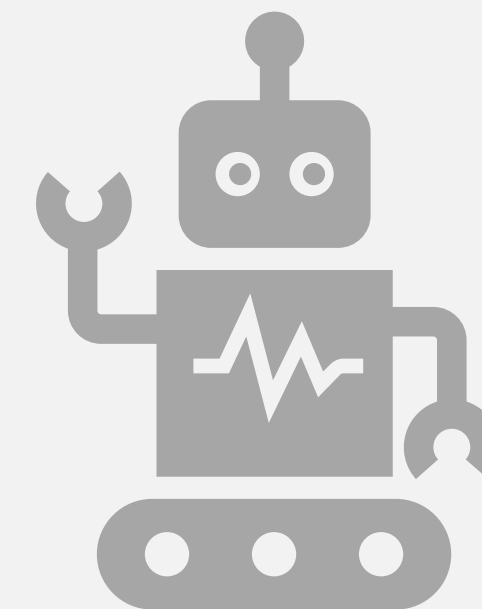


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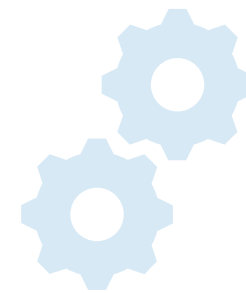
Anatomy of a Good Description



What it does



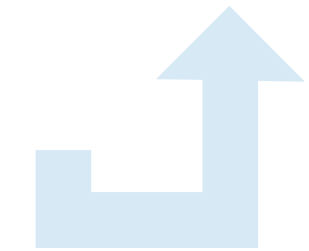
When to use it



Valid param values



Param units and format



Return value expectation

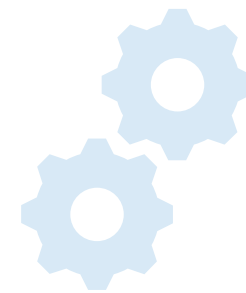
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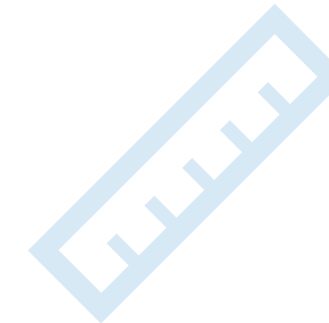
What it does



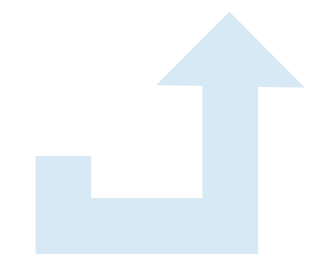
When to use it



Valid param values



Param units and format



Return value expectation

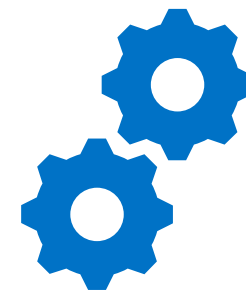
Anatomy of a Good Description



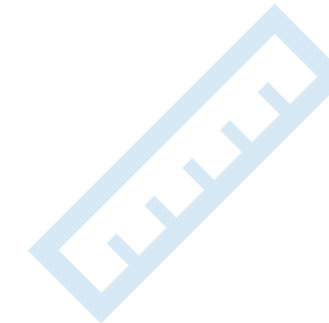
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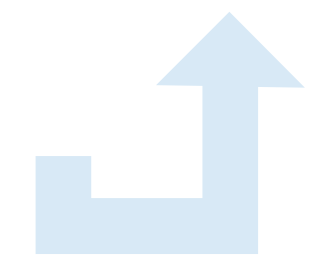
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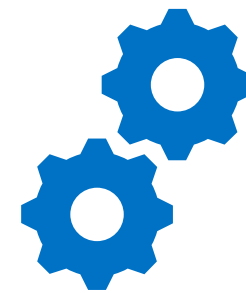
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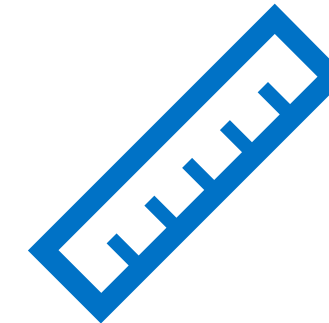
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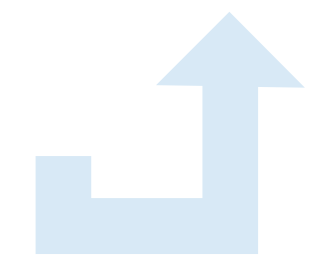
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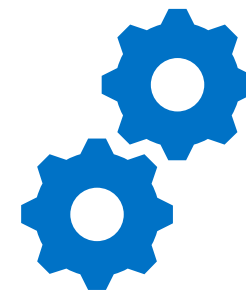
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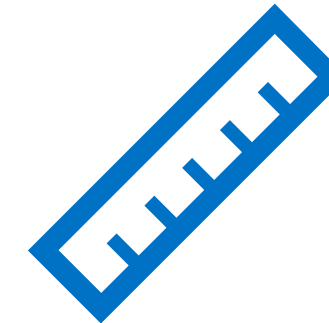
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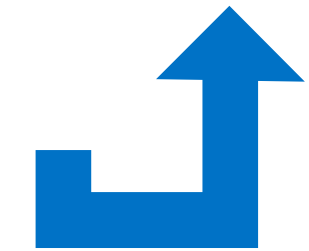
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public string GetTodos(bool? complete)
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Missing

Good Description

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What it does;
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Example param values

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Debugging MCP

MCP Inspector

- ✗ curl
- ✗ OpenAPI / Swagger
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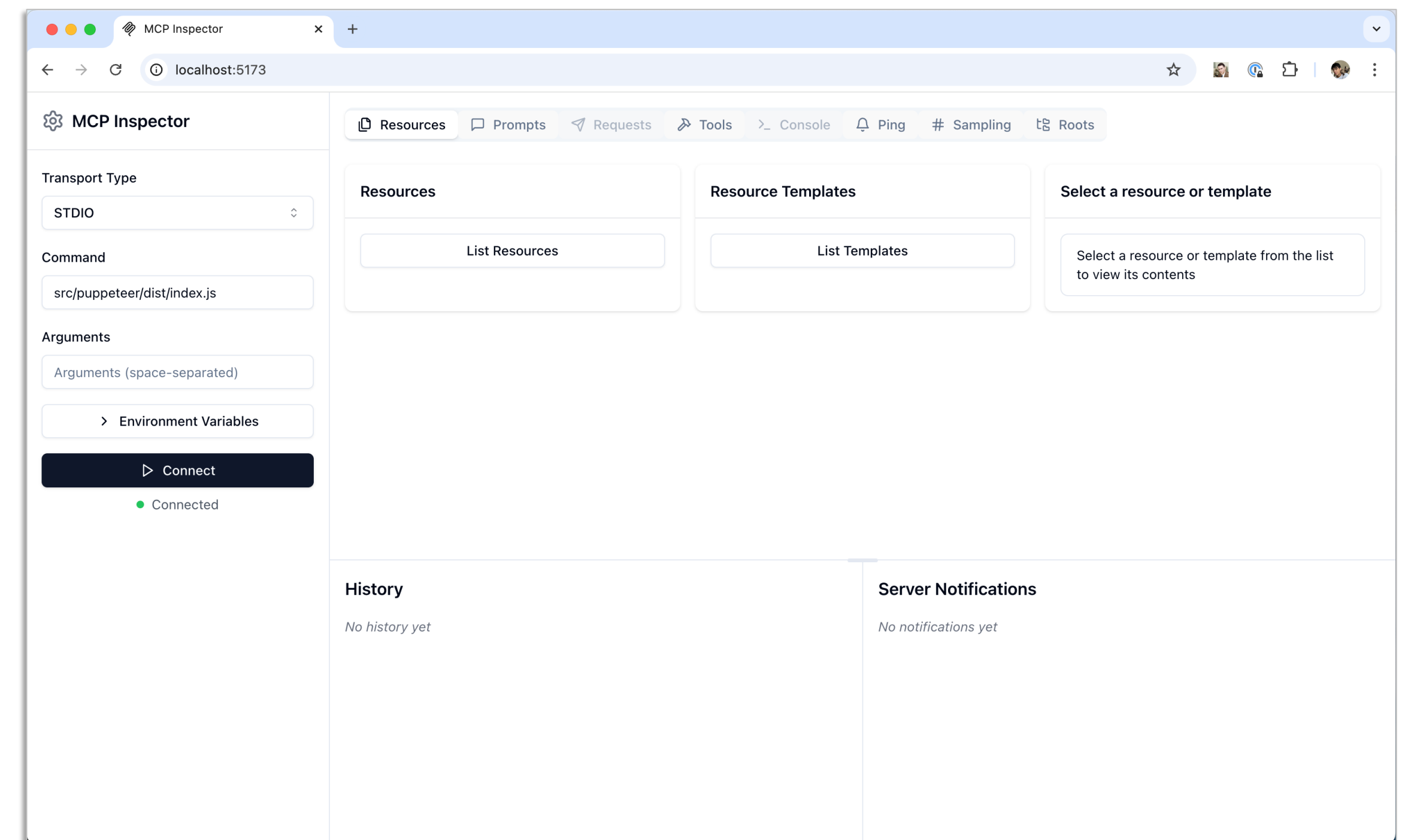
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DEMO

Security

From "Trust Me" to Enterprise-Grade



TRAILHEAD
TECHNOLOGY PARTNERS

Early MCP “Security”

- Every server **rolled its own**
- API keys in config files; stdio and localhost as the “security model”
- Fine for demos and local dev, but a non-starter for enterprise rollouts



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Now: Enterprise-Grade Security in MCP

- **OAuth 2.1** authorization is part of the MCP spec
- User identity flows through to downstream APIs
- Audience-bound tokens and security hardening baked into the SDK
- Stateless HTTP means gateways, WAF, APIM just work

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1. MCP **can be bolted on** to existing REST APIs
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3. Audience for REST is devs; for MCP it's AIs
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Thanks! Questions?

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- VP of .NET Foundation
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- trailheadtechnology.com/blog
- [jtowermi](https://twitter.com/jtowermi)
- Jonathan "J." Tower

LET'S TALK



tinyurl.com/th-offer

github.com/trailheadtechnology/dotnet-mcp-rest